2.2 Project Plan

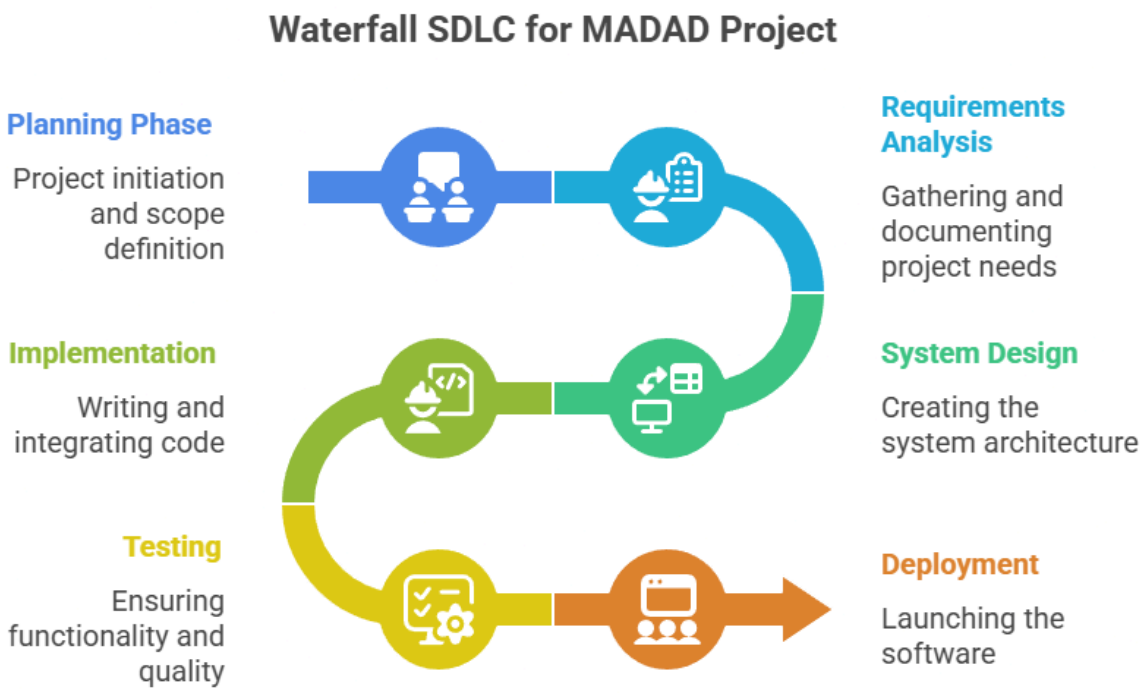
2.2.1Software Development Life Cycle

We looked at many methods, like Waterfall, Agile, Incremental, and Iterative models, and we find that The **Waterfall SDLC** is the best choice for MADAD based on the project's well-defined requirements, academic context, and structured two-part development process (gp1/ gp2) .

Waterfall is a linear, sequential methodology where each phase (Planning, Requirements Analysis, System Design, Implementation, Testing, Deployment) is completed before proceeding to the next.

Benefit:

- Waterfall's sequential phases align with the academic requirement , evaluated deliverables, ensuring advisor approval at each stage
- Reduces complexity compared to Agile's iterative sprints.
- Waterfall's linear process simplifies task allocation, with each team member focusing on one phase at a time.



2.2.2 Work Breakdown Structure (WBS)

WBS Development Approach Used: Analogy

The WBS for MADAD was developed using an **analogy-based approach**, where we referenced the task structure, durations, and milestones from similar past projects with comparable scope and complexity. This method helped estimate task durations, dependencies, and milestones more accurately without starting from scratch. It ensured realistic scheduling and alignment with proven project timelines, leading to a well-organized breakdown of activities—from team selection and planning to software implementation and final delivery—reflecting practical, experience-based insights.

Task ID | Name	Start Date	End Date	Duration
1 Team Selection	April 20, 2025	April 29, 2025	9 days
2 Idea Brainstorming	October 1, 2025	October 13, 2025	13 days
3 Idea Selection	October 14, 2025	October 15, 2025	2 days
4 Supervisor Selection	September 23, 2025	September 23, 2025	1 day
5 Idea Approval	October 15, 2025	October 16, 2025	2 days
6 Planning	October 16, 2025	October 18, 2025	3 days
7 Overview	October 19, 2025	October 21, 2025	3 days
8 Problem Statement	October 20, 2025	October 22, 2025	3 days
9 Problem Motivation	October 21, 2025	October 23, 2025	3 days
10 Similar Applications	October 22, 2025	October 27, 2025	6 days
11 Gap Analysis	October 24, 2025	October 28, 2025	5 days
12 Storyboard	October 27, 2025	November 3, 2025	8 days
13 Targeted Audience	November 3, 2025	November 5, 2025	3 days
14 Project Charter	November 5, 2025	November 10, 2025	6 days
15 Project Scope Statement	November 6, 2025	November 11, 2025	6 days
16 Assumptions and Constraints	November 7, 2025	November 10, 2025	4 days
17 Budget Summary	November 10, 2025	November 12, 2025	3 days
18 SDLC	November 11, 2025	November 14, 2025	4 days
19 WBS	November 12, 2025	November 18, 2025	7 days
20 Gantt Chart	November 13, 2025	November 19, 2025	7 days
21 Tools and Techniques Used	November 14, 2025	November 20, 2025	7 days
22 Risk Management Plan	November 18, 2025	November 19, 2025	2 days

Task ID | Name	Start Date	End Date	Duration
23 Prepare Interview	November 20, 2025	November 24, 2025	5 days
24 Prepare Questionnaire	November 21, 2025	November 25, 2025	5 days
25 Conduct Interview	November 26, 2025	November 27, 2025	2 days
26 Conduct Questionnaire	November 27, 2025	November 28, 2025	2 days
27 Analyze Interview Results	November 28, 2025	December 2, 2025	5 days
28 Collect and Analyze Questionnaire Results	December 1, 2025	December 4, 2025	4 days
29 Product Scope	December 3, 2025	December 8, 2025	6 days
30 Requirements Elicitation	December 4, 2025	December 9, 2025	6 days
31 Requirements Documentation	December 10, 2025	December 15, 2025	6 days
32 Requirements Management	December 12, 2025	December 17, 2025	6 days
33 Context Diagram	December 18, 2025	December 29, 2025	12 days
34 Activity Diagram	December 20, 2025	December 31, 2025	12 days
35 Sequence Diagram	December 22, 2025	January 5, 2025	15 days
36 Graduation Project 1 Documentation Completion	January 6, 2025	January 31, 2025	26 days

Task ID | Name	Start Date	End Date	Duration
37 Software Implementation	February 15, 2025	March 31, 2025	45 days
38 Front-End Development	February 15, 2025	March 10, 2025	24 days
39 Back-End Development	March 11, 2025	March 28, 2025	18 days
40 Database Development	March 29, 2025	April 3, 2025	6 days
41 Database Integration	April 4, 2025	April 9, 2025	6 days
42 Software Testing	April 10, 2025	April 16, 2025	7 days
43 Create Test Plan	April 10, 2025	April 10, 2025	1 day
44 Unit Testing	April 11, 2025	April 14, 2025	4 days
45 Integration Testing	April 15, 2025	April 16, 2025	2 days
46 Performance Testing	April 15, 2025	April 16, 2025	2 days
47 Security Testing	April 17, 2025	April 17, 2025	1 day
48 Software Deployment	April 18, 2025	April 22, 2025	5 days
49 Execute Deployment Plan	April 18, 2025	April 21, 2025	4 days
50 Project Delivery	April 22, 2025	June 13, 2025	53 days